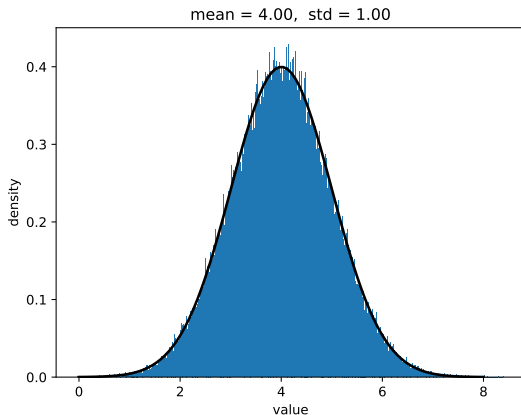


Machine learning Tek 5: overview of mathematical tools



Probabilities and statistics

Optimization

Metrics

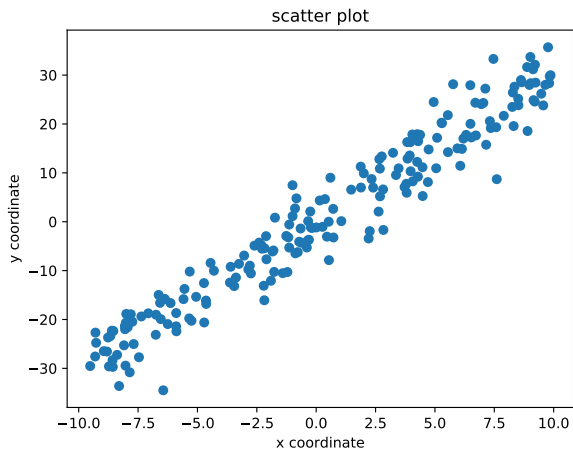
- Metrics in output space

- Metrics in input space

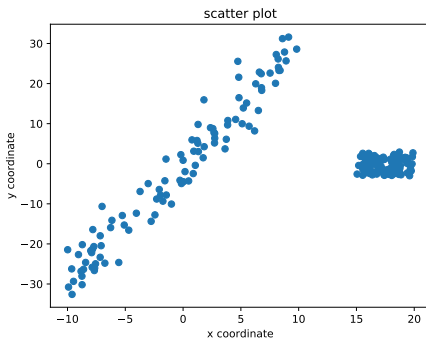
- Outliers

To have a solid understanding of machine learning, it is necessary to be familiar with elementary probabilities and statistics.

Random variables



Random variables



We want to analyse how the data are **distributed**. For instance the x coordinate, the y coordinate.

Random variables

- ▶ (informal definition) A **random variable** is a quantity that can take several values, with some randomness.
- ▶ https://en.wikipedia.org/wiki/Random_variable

Random variables

- ▶ A **random variable** is a quantity that can take several values
- ▶ For instance :
 - ▶ the result of a dice throw

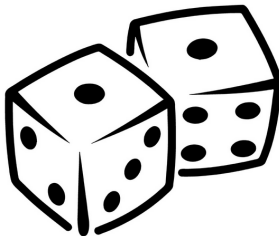


Figure – Dice

Random variables

- ▶ A **random variable** is a quantity that can take several values
- ▶ For instance :
 - ▶ the result of a dice throw
 - ▶ waiting time with RATP



Figure – Some metro station

Random variables

- ▶ A **random variable** is a quantity that can take several values
- ▶ For instance :
 - ▶ the result of a dice throw
 - ▶ waiting time with RATP
 - ▶ weather



Figure – Weather in November

Random variables

- ▶ A **random variable** is a quantity that can take several values
- ▶ For instance :
 - ▶ the result of a dice throw
 - ▶ waiting time with RATP
 - ▶ weather
 - ▶ number of cars taking the périphérique at the same time

Why are random variables important ?

- ▶ most datasets encountered in machine learning can be considered as sampled from random variables.
- ▶ this is important for theoretical studies, and hence for applications : a better theoretical understanding of a problem allows to choose the best algorithm to solve it.
- ▶ theoretical results are sometimes precise in the sense that they allow to estimate the order of magnitude of the statistical error (e.g. the prediction error) as a function of d (dimension of the samples) and n (number of samples)
- ▶ a subdomain of machine learning is "statistical learning"

Random variables

- ▶ Some are random variables **continuous**, others **discrete**
- ▶ **continuous** : weather, RATP
- ▶ **discrete** : dice (6 possibilities), number of cars (> 10000)

Probability distributions

- ▶ A random variable is linked to a **probability distribution**, which is a function P
- ▶ It quantifies the probability of observing one outcome.
- ▶ For a discrete variable : each possible outcome is associated with a number between 0 and 1

Probability distributions

- ▶ For a dice game, the possible outcomes are in the set $\{1, 2, 3, 4, 5, 6\}$
- ▶ For a dice game : $P(1) = ?$ $P(2) = ?$ $P(3) = ?$ $P(4) = ?$
 $P(5) = ?$ $P(6) = ?$

Probability distributions

- ▶ For a dice game, the possible outcomes are in the set $\{1, 2, 3, 4, 5, 6\}$
- ▶ For a dice game : $P(1) = \frac{1}{6}$, $P(2) = \frac{1}{6}$, $P(3) = \frac{1}{6}$, $P(4) = \frac{1}{6}$, $P(5) = \frac{1}{6}$, $P(6) = \frac{1}{6}$
- ▶ This is called a **uniform distribution**

Probability distributions

- ▶ Périphérique : probably a time-dependent very complicated distribution

Continuous variables

- ▶ The situation is different for continuous random variables.
- ▶ The distribution is given by a **probability density function**. Informally, the probability of being between x and $x + dx$ is $p(x)dx$.
- ▶ https://en.wikipedia.org/wiki/Probability_density_function
- ▶ Note that some variables are neither discrete nor continuous.

Uniform discrete

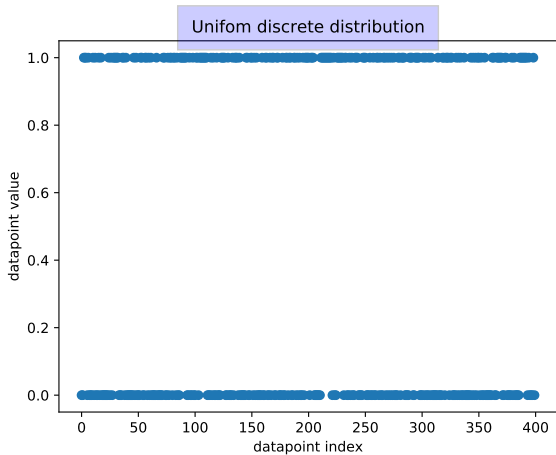


Figure – Uniform discrete distribution with 2 values

Uniform discrete

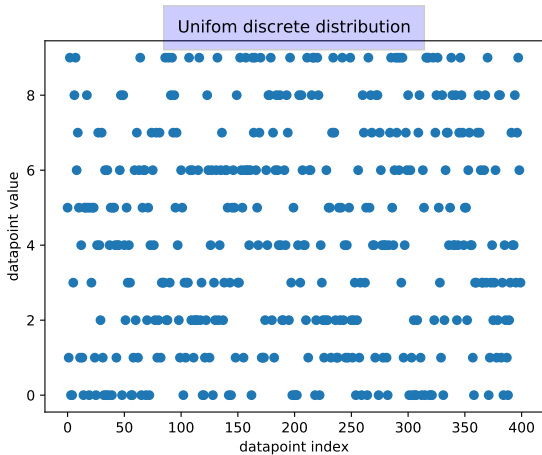


Figure – Uniform discrete distribution with 10 values

Bernoulli

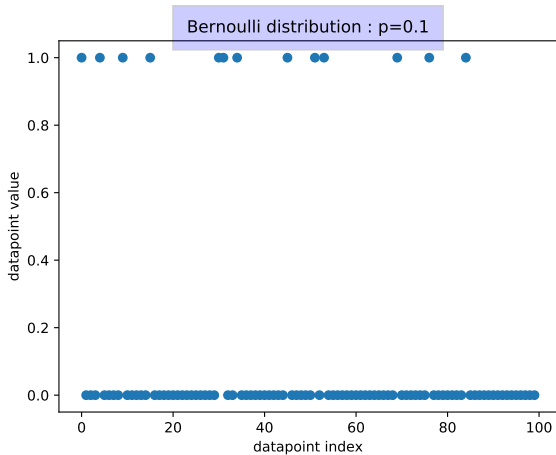


Figure – Bernoulli distribution

Bernoulli p

- ▶ With probability p , $X = 1$
- ▶ With probability $1 - p$, $X = 0$

Bernoulli

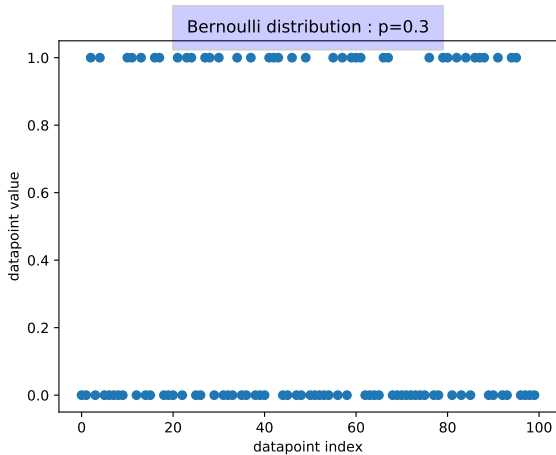


Figure – Bernoulli Distribution

Bernoulli

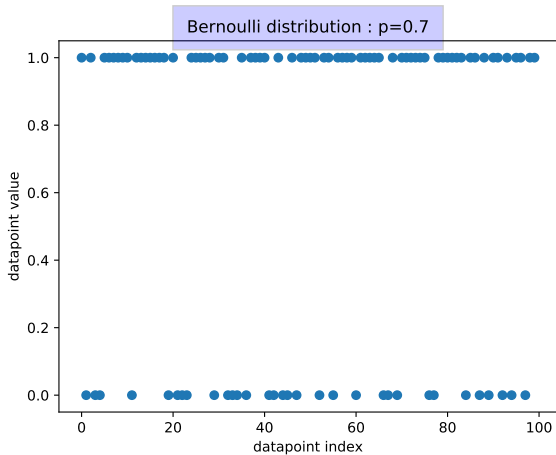


Figure – Bernoulli Distribution

Uniform continuous

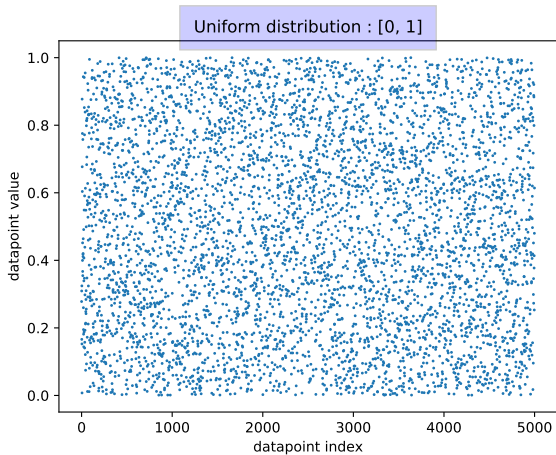


Figure – Uniform continuous distribution

Uniform continuous

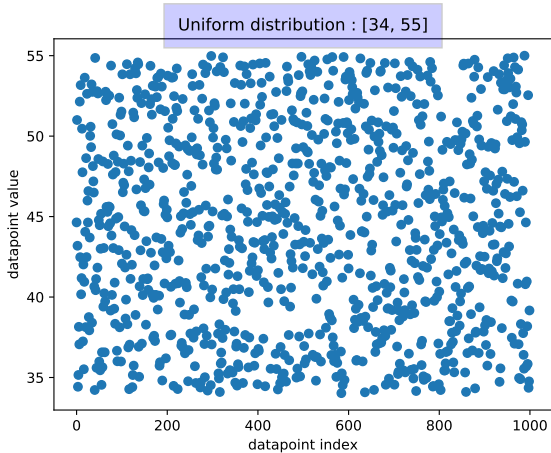


Figure – Uniform continuous distribution

Uniform continuous

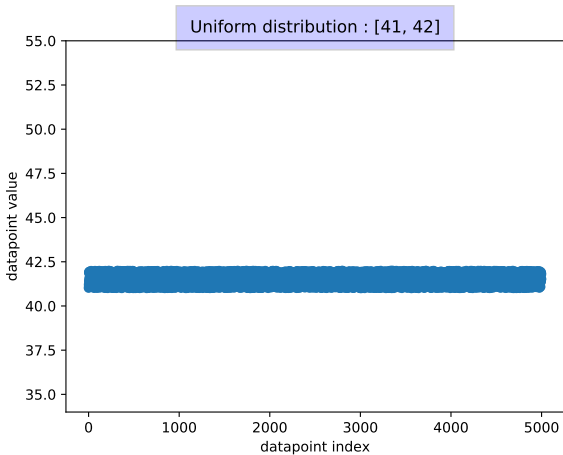


Figure – Uniform continuous distribution

Normal

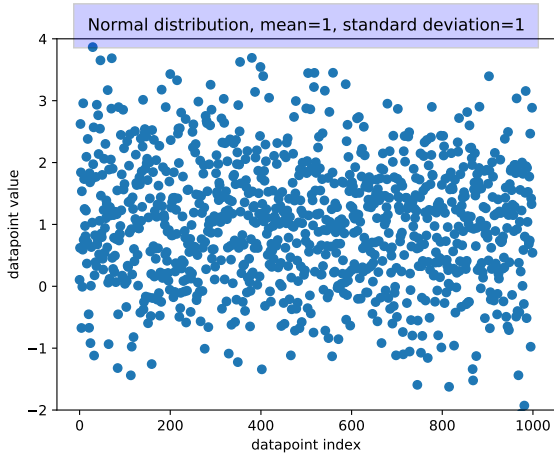


Figure – Normal distribution

Normal

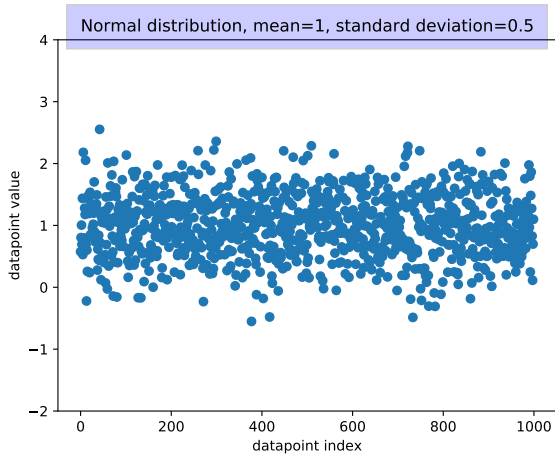


Figure – Normal distribution

Normal

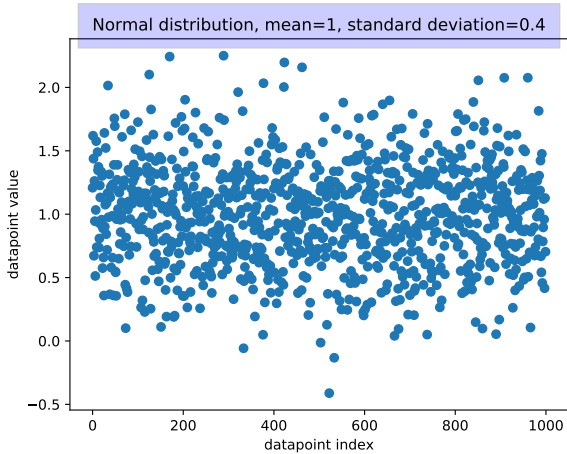


Figure – Normal distribution

White noise

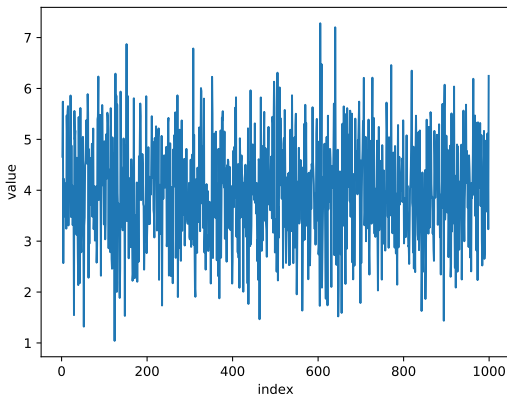


Figure – White noise

Histograms

Histograms are an alternative representation of the results of a (one-dimensional) random variable.

Uniform discrete

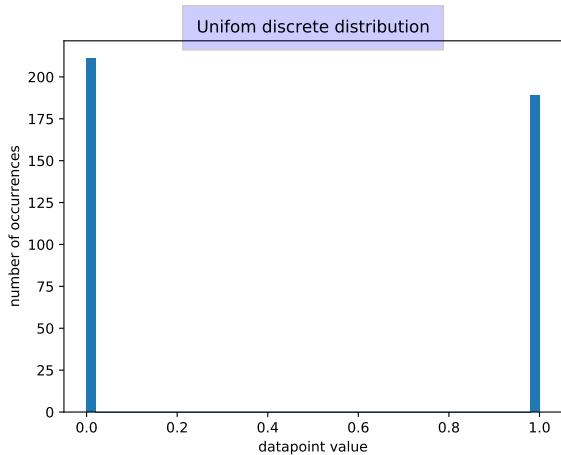


Figure – Histogram 1

Uniform discrete

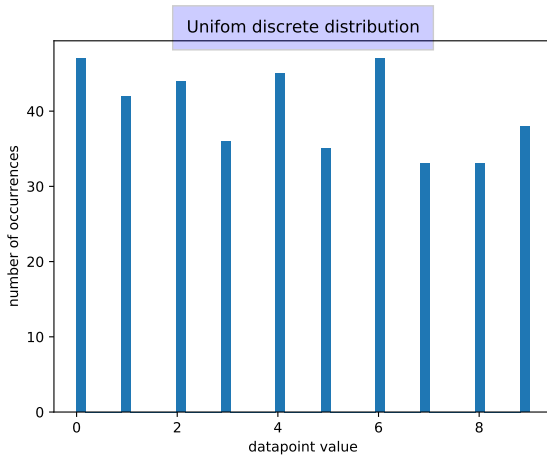


Figure – Histogram 1

Bernoulli

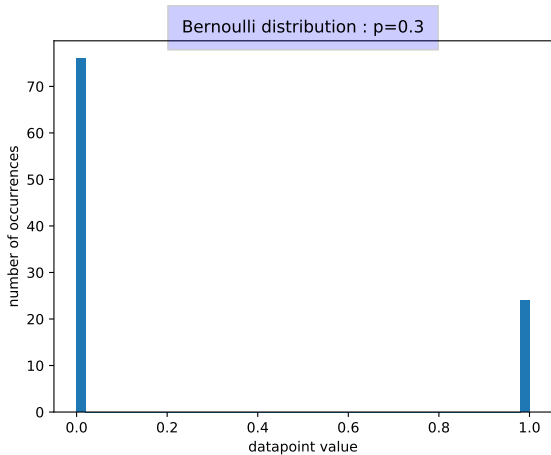


Figure – Histogram 2

Uniform continuous

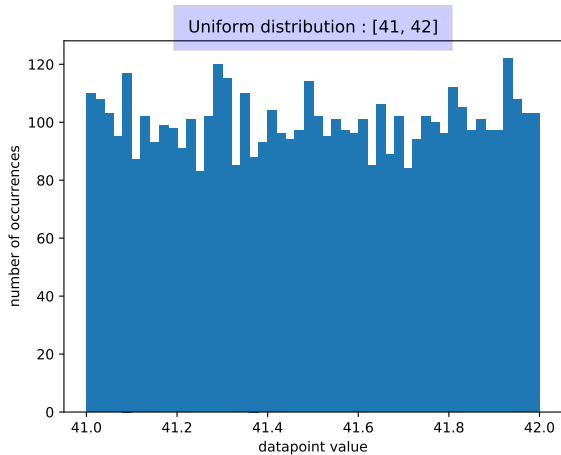


Figure – Histogram 3

Normal

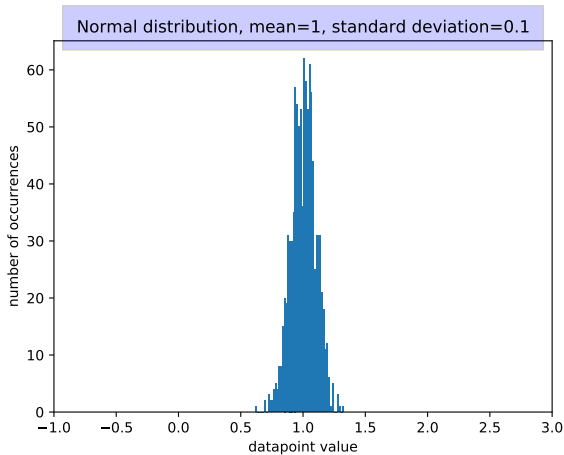


Figure – Histogram 4

Normal

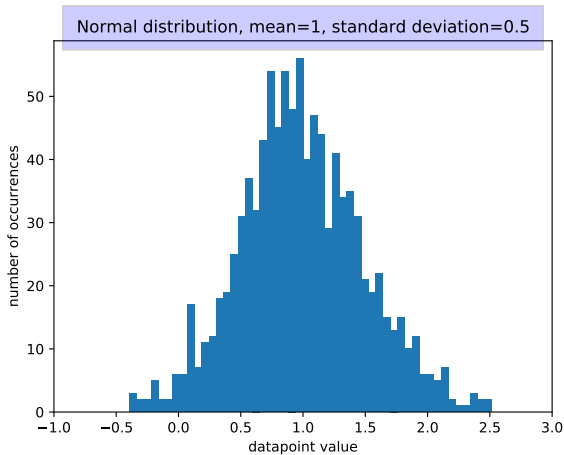


Figure – Histogram 4

cd distributions/

We can use the files **analyze_distribution_1.py** and **analyze_distribution_2.py** to analyze and plot some simple datasets, stored in **csv_files/**

Distribution 1

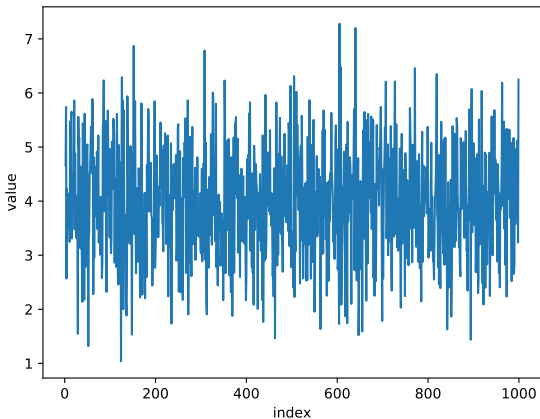


Figure – The data we analyze

histograms

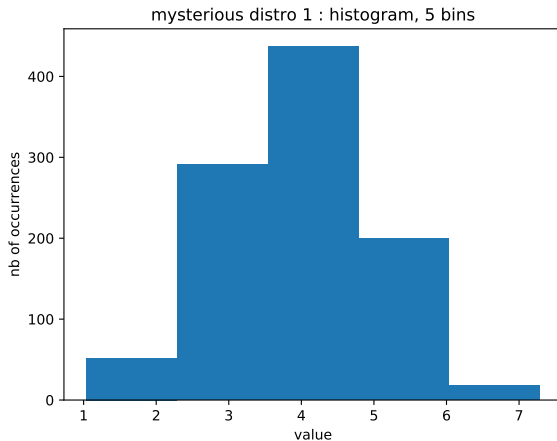


Figure – 5 bins

histograms

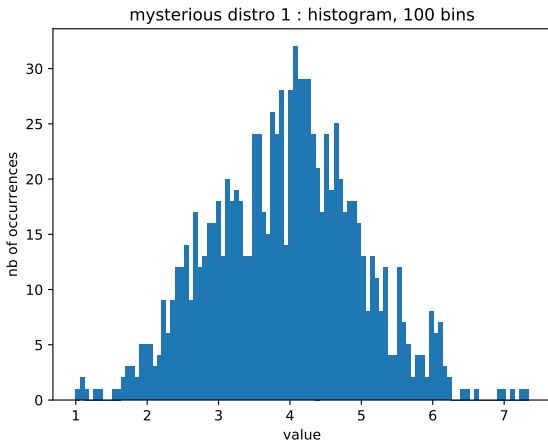


Figure – 100 bins

histograms

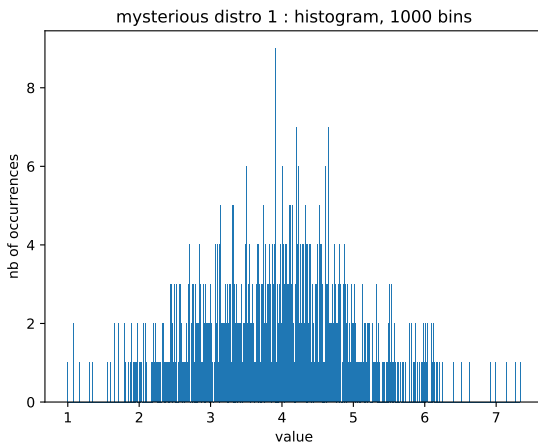


Figure – 1000 bins (too many)

Normal distribution

```
import csv
import numpy as np

file_name = 'mysterious_distro_1.csv'

mean = 4
std_dev = 1
nb_point = 1000

with open('csv_files/' + file_name, 'w') as csvfile:
    filewriter = csv.writer(csvfile, delimiter=',')
    for point in range(1, nb_point):
        random_variable = np.random.normal(loc=mean, scale=std_dev)
        filewriter.writerow([str(point), str(random_variable)])
```

Figure – `create_normal.py` : Creation of the distribution

Second example

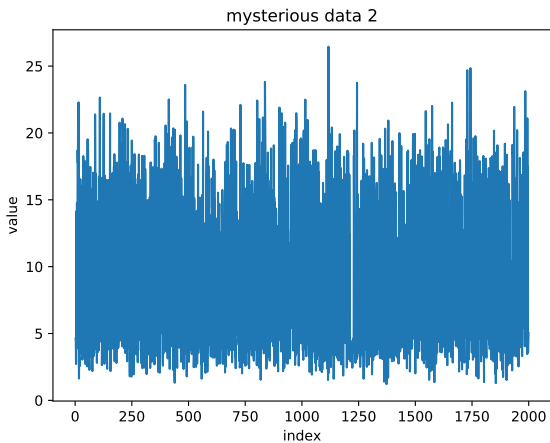


Figure – Second distribution

Exercise 1 : Create a one-dimensional dataset with a histogram that looks like this one !

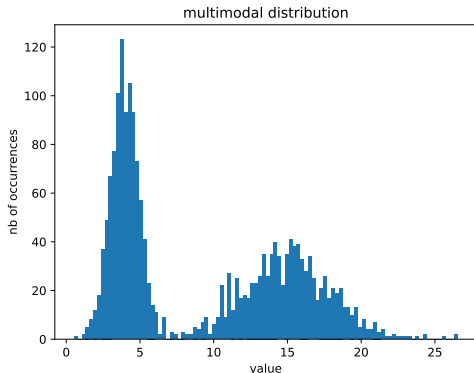


Figure – This distribution has several **modes**

Expected value

- ▶ The **expected value** of a random variable is its probabilistic average.
- ▶ Under the condition that this probabilistic average is correctly defined.

Expected value (espérance)

- ▶ For a discrete random variable X that takes the values x_i with probability p_i :

$$E(X) = \sum_{i=1}^n p_i x_i \quad (1)$$

- ▶ For a continuous random variable X with density p :

$$E(X) = \int x p(x) dx \quad (2)$$

Note that X may have values in $\mathbb{R}^d$, with $d \geq 1$.

Expected value (espérance)

Exercice 2 : Computing an expected value

- For a discrete random variable X that takes the values x_i with probability p_i :

$$E(X) = \sum_{i=1}^n p_i x_i \quad (3)$$

- For a continuous random variable X with density :

$$E(X) = \int x p(x) dx \quad (4)$$

Compute the expected value of the dice game.

Variance

The variance is a measure of the dispersion of a random real variable.

<https://en.wikipedia.org/wiki/Variance>

$$\text{var}(X) = E\left((X - E(X))^2\right) \quad (5)$$

Note that we can also define the variance of a multidimensional random variable (which means a random vector). In that case, it is a matrix.

Multidimensional vectors

We often consider random variables and data that live in spaces with a higher dimension than 2 (random vectors).

- ▶ images
- ▶ sensor that receives **multimodal information**

Correlation

Random vectors with correlated components are common statistical objects.

- ▶ In physics, temperature and pressure, measured by some sensors are correlated.
- ▶ In a dataset of customers of a company, some dimensions are likely to be correlated.

To study the statistical relationship between components, we can compute the **covariance** of the two components, or the **correlation**, (normalized covariance (see below)).

<https://en.wikipedia.org/wiki/Correlation>

Covariance

The covariance is a measure of the relationship between the variations of two random variables.

$$\text{cov}(X, Y) = E\left((X - E(X))(Y - E(Y))\right) \quad (6)$$

Correlation

The correlation is the covariance divided by the square roots of the variances.

$$\text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} \quad (7)$$

Example

The data in `csv_files/distribution_3.csv` contain samples of a random variable with 5 dimensions (random vector). Some of these dimensions are correlated.

Covariance

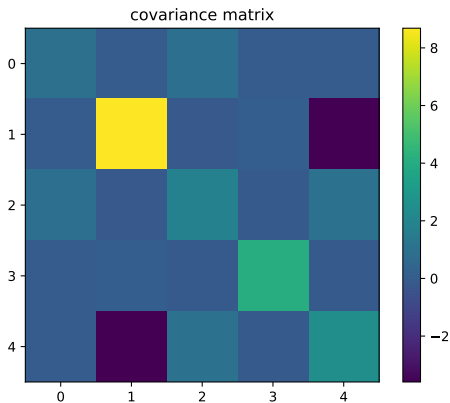


Figure – Covariance matrix of the random vector.

Correlation matrix

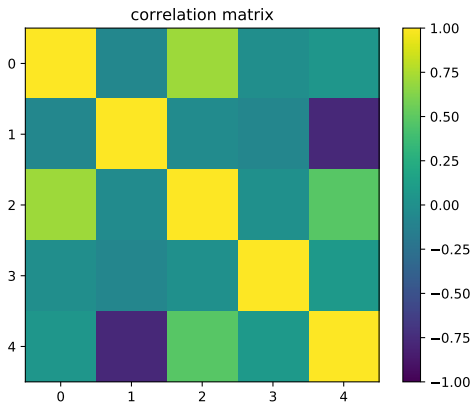


Figure – Correlation matrix for the distribution, note the difference in the scale.

Pandas, scikit-learn

- ▶ <https://pandas.pydata.org/>
- ▶ https://scikit-learn.org/stable/datasets/toy_dataset.html
- ▶ pandas demo with iris and distribution 3.

Minimization of a function

Optimization is another core aspect of machine learning.

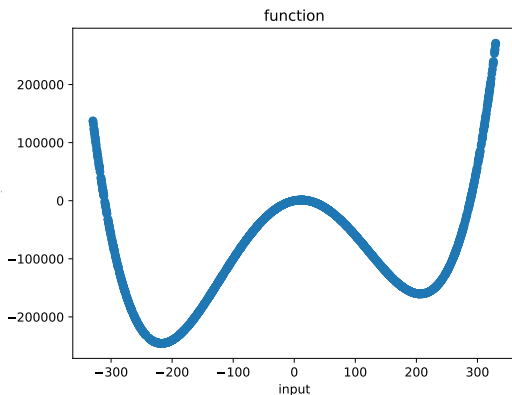


Figure – Loss function

Optimization in machine learning

The loss function typically represents the quality of a set of parameters to solve a problem.

- ▶ in supervised learning, typically a measure of the prediction error on the dataset
- ▶ in clustering, typically a distortion
- ▶ in density estimation, a likelihood

Analytic minimization

Exercise 3: What is the minimum of the function

$$f : x \rightarrow (x - 1)^2 + 3.5 \quad (8)$$

And for what value x is it obtained ?

Iterative algorithms

However, in most applications of machine learning, it is not possible to use an analytical solution, either because :

- ▶ we do not know the analytical solution
- ▶ we know how to compute it, but the computation is too costly for practical use.

Instead, we use **iterative algorithms** (gradient descent, coordinate descent, etc.)

Gradient algorithms

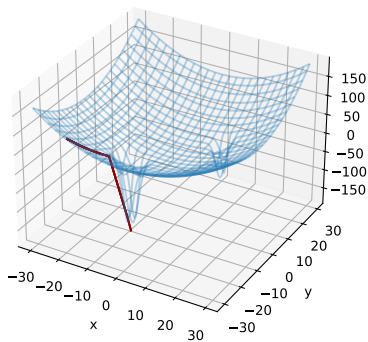


Figure – Optimization trajectory.

Metrics

Let $D = \{x_1, \dots, x_n\} \subset \mathcal{X}$ be a dataset of n samples, with labels $\{y_1, \dots, y_n\} \subset \mathcal{Y}$.

There is a metric in the input space $\mathcal{X}$ and in the output space $\mathcal{Y}$.

- ▶ The **metric** in $\mathcal{X}$ determines to what extent two samples x_i and x_j should be considered similar or dissimilar.
- ▶ The **metric** in $\mathcal{Y}$ determines to what extent two labels y_i and y_j should be considered similar or dissimilar.

In all machine learning, the choice of the metric is very important !

Metrics in output space

A **loss function** l is a map that measures the discrepancy between to elements of a set (for instance of a linear space).

$$l : \begin{cases} \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+ \\ (y, z) \mapsto l(y, z) \end{cases}$$

Typically, z can represent our prediction for a given input x , $z = \tilde{f}(x)$, and y the correct label.

Most common supervised learning losses

"0-1" loss for binary classification.

$\mathcal{Y} = \{0, 1\}$ or $\mathcal{Y} = \{-1, 1\}$.

$$l(y, z) = 1_{y \neq z} \quad (9)$$

Squared loss for regression

$\mathcal{Y} = \mathbb{R}$.

$$l(y, z) = (y - z)^2 \quad (10)$$

absolute loss for regression.

$\mathcal{Y} = \mathbb{R}$.

$$l(y, z) = |y - z| \quad (11)$$

In **unsupervised learning**, there is notion of **output space** !

Geometric distances

Often, $\mathcal{X} = \mathbb{R}^p$ (input space). In this case, **geometric** metrics are used. $x = (x_1, \dots, x_p)$ and $y = (y_1, \dots, y_p)$ are p -dimensional **vectors**.

- ▶ $L_2 : \|x - y\|_2 = \sqrt{\sum_{k=1}^p (x_k - y_k)^2}$ (Euclidian distance, 2-norm distance)
- ▶ $L_1 : \|x - y\|_1 = \sum_{k=1}^p |x_k - y_k|$ (Manhattan distance, 1-norm distance)
- ▶ weighted $L_1 : \sum_{k=1}^p w_k |x_k - y_k|$, with each $w_k > 0$.
- ▶ $\|x - y\|_\infty : \max(|x_i - y_i|, i \in [1, n])$ (infinity norm distance, Chebyshev distance)

Choice of the metric

In some contexts, some usual metrics such as $L2$ might not be meaningful !

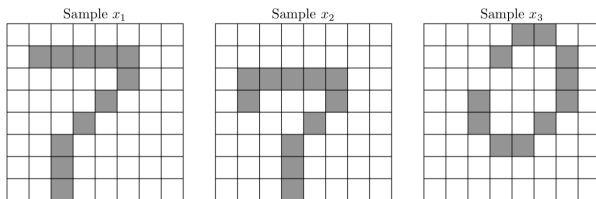


Figure – In $\mathbb{R}^{64}$, those three points form an equilateral triangle, [Fix et al., ,]

Non-geometric data

Not all data are geometric !

Hamming distance

- ▶ $\#\{x_i \neq y_i\}$ (Hamming distance)
- ▶ Levenshtein distance for strings (allows deletions and additions)

General definition of a distance

A **distance** on a set E is an application $d : E \times E \rightarrow \mathbb{R}_+$ that must :

- ▶ be **symetric** : $\forall x, y, d(x, y) = d(y, x)$
- ▶ **separate the values** : $\forall x, y, d(x, y) = 0 \Leftrightarrow x = y$
- ▶ respect the **triangular inequality**
 $\forall x, y, z, d(x, y) \leq d(x, z) + d(y, z)$

General definition of a distance

We could verify that :

- ▶ L2 is a distance
- ▶ Hamming is a distance

Similarities

Sometimes, it is not possible to define a proper **distance** in the input space $\mathcal{X}$! This may happen for instance if $\mathcal{X}$ is a dataset of texts.

- ▶ When distances are unavailable, we can use **Similarities** or **Dissimilarity** to compare points.
- ▶ Dissimilarities are more general and don't always abide by the distance axioms.
- ▶ Other examples : Adjacency in an oriented graph, Custom aggregated score to compare data.

Example : cosine similarity

The **cosine similarity** may be used to compare texts.

If u and v are vectors,

$$S_C(u, v) = \frac{(u|v)}{||u|| ||v||} \quad (12)$$

- ▶ the **bag of words representation** allows us to build a vector from a text (one hot encoding).
- ▶ `cosine_similarity/scrapper.py`
- ▶ `cosine_similarity/similarity.py`

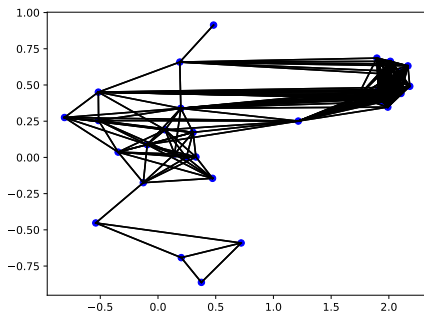
Hybrid data

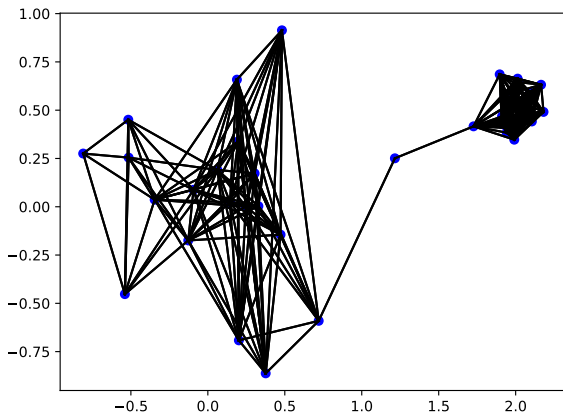
Sometimes each sample contains both numerical data and non-numerical data (text, categorical data.)

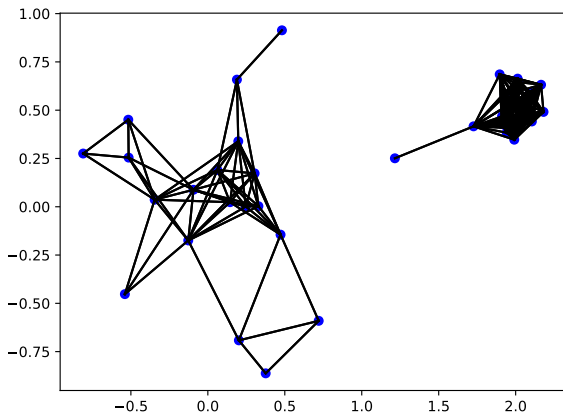
See **day_1/4_metrics/hybrid_data/**

This is often the case in machine learning applications! (database of customers, cars, countries, etc.)

Exercise 4: Using the notebook in `day_1/4_metrics/geometric_data/`, choose the metric and the threshold so that this graph (and the ones on the next slides) are built.

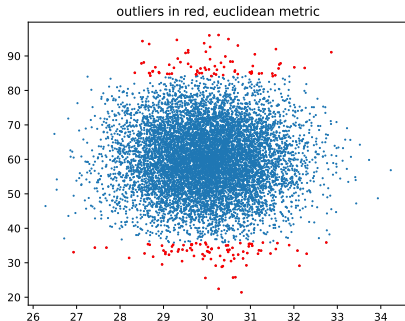






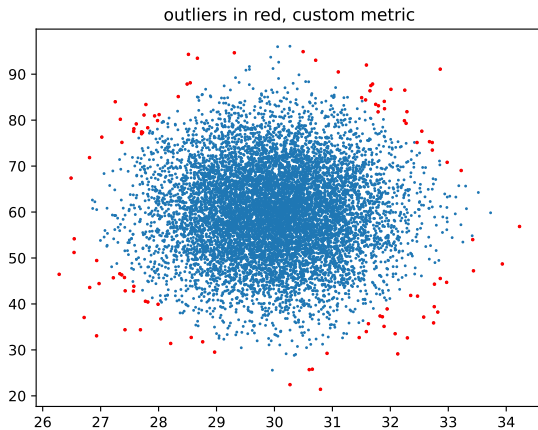
Outliers

Outliers are samples that are considered like non-representative of the dataset (e.g. due to a measurement error). The detection of outliers depends on the metric !



Figure

Outliers with a different metric



References I



Fix, J., Frezza-Buet, H., Geist, M., and Pennerath, F.
Machine Learning.pdf.